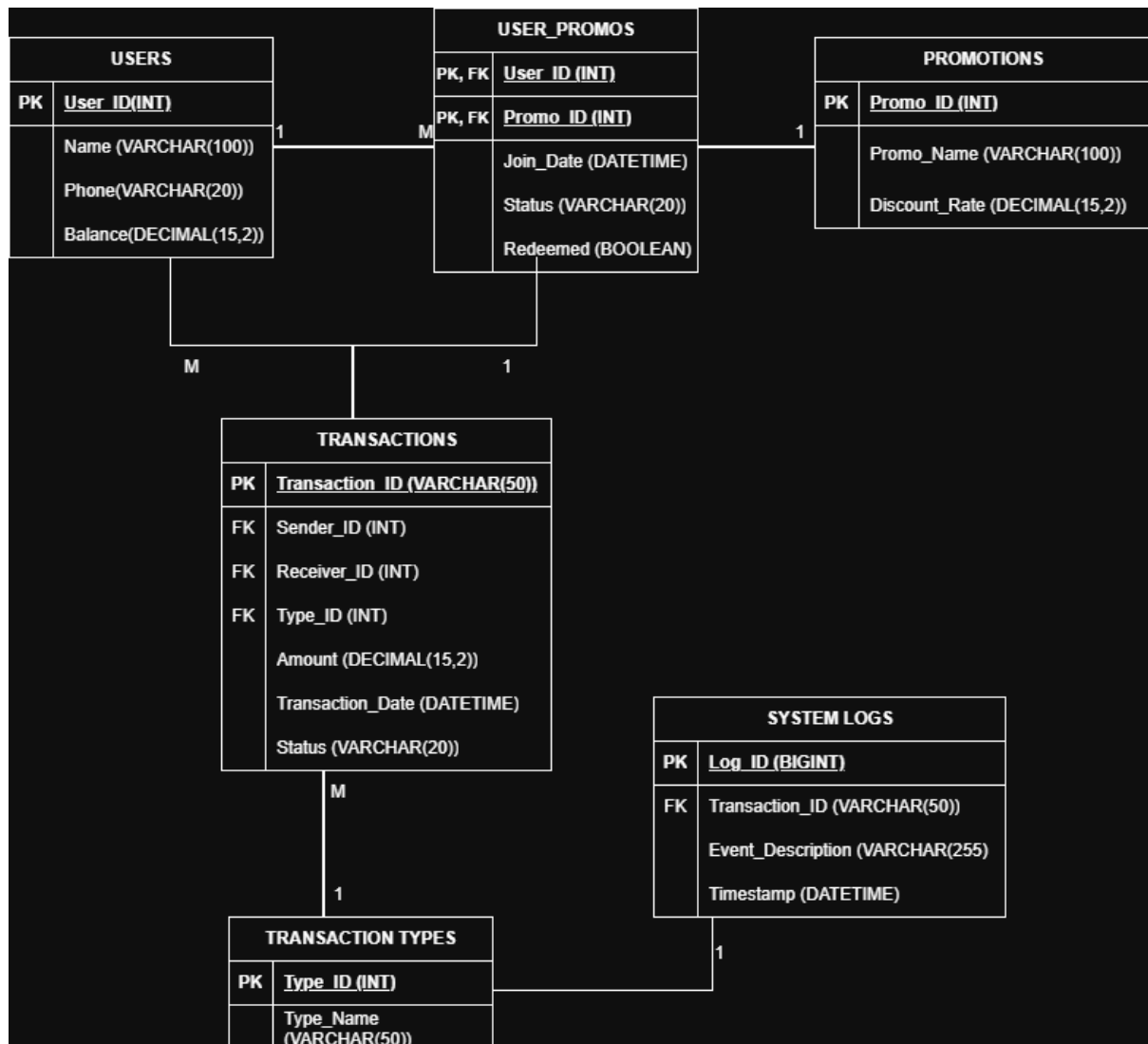




EDR DOCUMENTATION

This Entity Relationship Diagram(ERD) transforms data from the MoMo SMS data processing system which is hierarchical into a normalized schema which is suitable for the high data processing in the files.

The first and central design was the Users table, which is primary throughout the data. In this table, data is separated by use of attributes like name and phone number which are user specific. The transaction table is also a primary table in the diagram. In this design the transactional table links to the Users table through foreign key attributes like Sender_ID and Receiver_ID. This shows the close relationships in Mobile Money where a single user can both be a sender and a recipient of money.

The Transaction Types table is for standardisation of activities in the transactions, like "Payment", "Received" and "Bank deposits" which are found in the Momo files. This helps in efficiency when making financial reports and proper organisation of the types of activities happening in the transaction history.

The User_Promos junction table was created to satisfy the many to many relationship in the data between the User and Promotions. This entity tracks unique promotion instances like the “BivaMoMotima” campaign, hence capturing attributes like “status” and “Join_Date” which cannot be found in either tables.

Finally, the System_Logs table is important for proper system monitoring. This table links logs to specific Transactional_IDs, hence ensuring traceability in all the transactions on the XML source file.